

Capstone Project -- Pressure Forecasting -- Report

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1. Question and Motivation

The question is whether production and injection data can reliably forecast next month's reservoir pressure at the Volve field. Pressure decline is a signal that injection or withdrawal rates need some adjustment. Therefore, a dependable one-month forecast would be useful to production and reservoir engineers' decision making of when to change water injection rates. The test is whether a model can beat the simplest possible forecast: the persistence (naïve) baseline. The naïve baseline is simply that next month's pressure equals this month's. The persistence baseline scores an RMSE 85.3 psia on the last 18 months that were held out. A useful model is one that clearly improves on that.

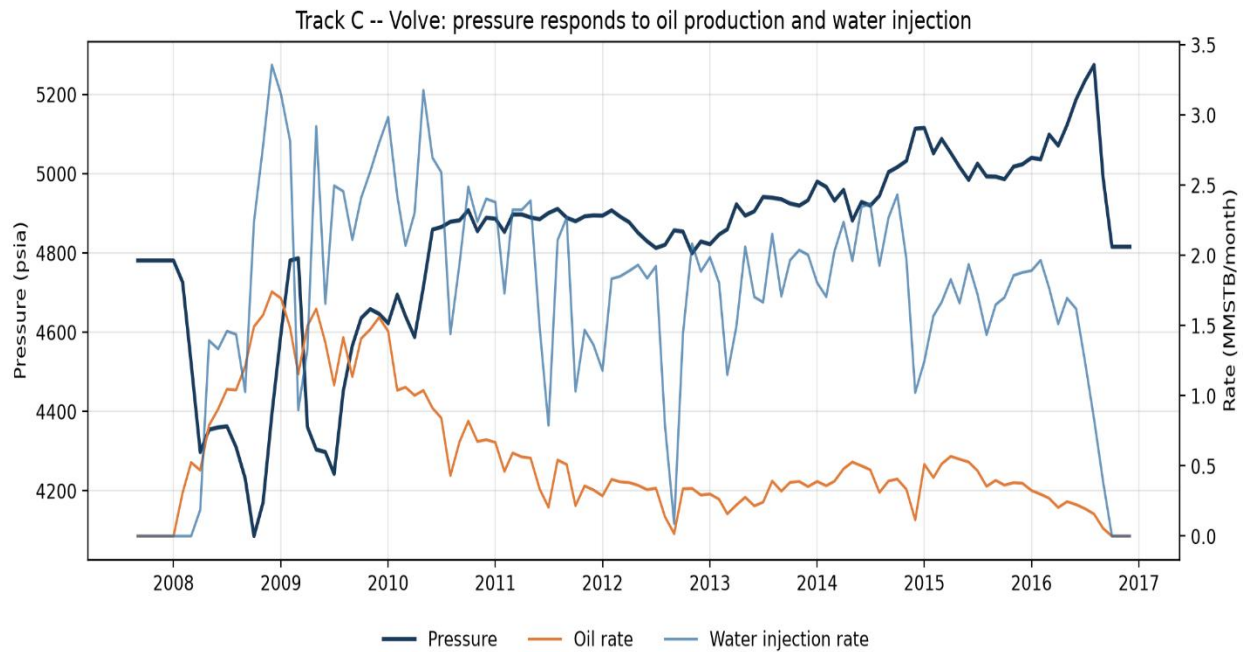
2. Data

The dataset is a monthly field-level summary of Volve, with eight columns: average reservoir pressure, cumulative oil, gas and water production, gas and water injection, and gas-oil ratio. It comes from a public, community-processed version (yohanesnuwara/pyreservoir on GitHub, <https://github.com/yohanesnuwara/pyreservoir>), itself built from Volve's Eclipse reservoir-simulation history-match output. It contains 112 months, from 2007-09-01 to 2016-12-01, with no missing values.

Cleaning was limited. The Date column was parsed to datetime and the rows sorted chronologically. Because the production and injection columns are cumulative, monthly oil rate and water injection rate were created by differencing them and converting from STB to MMSTB (the first month is set to zero). The models use pressure, oil rate and water injection rate; gas injection is zero throughout and was not used.

Known quality issues. These are simulation-derived field averages, not gauge measurements, so the series is smoother than raw pressure data and reflects the history match. Production starts in 2008-02 and water injection in 2008-04, so the first six months are flat pre-production data; they

serve only as lag history, which leaves 88 monthly training samples and 18 test samples. The record is a single field and only 112 points long.



3. Evaluation Design

The data were split chronologically. The first 88 monthly samples (2008-03 to 2015-06) were used for training and tuning, and the last 18 months (2015-07 to 2016-12) were held out for testing. A chronological split matches real use, where a forecast is made forward from known history. A shuffled split would let the model train on months adjacent to its test months, and since neighbouring months are highly correlated and lagged windows overlap, this would overstate accuracy. The first six months of the record only supply lag history, which is why 88 training samples remain from 94 months.

The primary metric is root-mean-square error (RMSE, in psia), which penalises the large misses that would trigger an operating decision; mean absolute error (MAE, psia) is reported alongside it as a measure of typical error. The baseline is persistence (next month's pressure equals this month's), which scores RMSE 85.3 psia and MAE 49.5 psia on the test months. A model is considered useful only if it beats persistence by a margin that survives resampling.

Leakage was controlled in four ways. Inputs are lagged values only, so no feature uses information from the month being forecast. Scalers were fitted on training months only (inside each fold for the LSTM). Hyperparameters and lag lengths were tuned with walk-forward cross-validation (TimeSeriesSplit) on the training months only. The test months were used to report results, not to choose between models.

4. Methods

The target is next month's pressure, modelled as the change from the current month.

$$\Delta P = P(t) - P(t-1)$$

The forecast is the last known pressure plus the predicted change. This makes persistence the special case $\Delta P = 0$ and avoids the extrapolation problem of predicting the pressure level directly.

Inputs are six months of lagged pressure changes, optionally with six months of lagged oil rate and water injection rate (MMSTB per month), standardised using training statistics. Three model families were built: a Random Forest on pressure level (baseline); Ridge regression on ΔP , with its regularisation strength chosen from $\{1, 10, 100, 1000, 10000\}$ by walk-forward CV; and an LSTM on ΔP (16 hidden units, learning rate 0.005, 300 epochs), whose predictions are the mean of five random seeds.

Four experiments were run, each stated as a hypothesis in the notebook: (1) predicting ΔP instead of the pressure level; (2) varying the history window to 3, 6 and 12 months, with and without drivers; (3) forecasting oil rate instead of pressure; and (4) recursive six-month-ahead forecasts. Within each model family, the setting used for the final comparison was chosen by CV on the training months, and the test result was read afterwards.

5. Results

Table 1 shows the one-step pressure models on the 18 held-out months. It is a condensed version of the full lag grid, which stays in the notebook and the results log.

Model	RMSE (psia)	MAE (psia)	RMSE vs persistence	Note

Persistence	85.3	49.5	0.0%	baseline
Random Forest, pressure lags only	92.8	68.1	-8.8%	
Random Forest, pressure + drivers	95.1	68.4	-11.5%	
Ridge on ΔP , pressure only (LAGS=6)	85.5	49.4	-0.2%	
Ridge on ΔP, pressure + drivers (LAGS=6)	82.0	50.3	+3.9%	best Ridge by CV
LSTM, pressure + drivers (LAGS=12)	90.7	72.3	-6.3%	best LSTM by CV
LSTM, pressure only (LAGS=3)	79.0	49.7	+7.4%	lowest test RMSE; n

Table 1 Every model's test RMSE is within roughly $\pm 10\%$ of persistence, and the best CV-selected models do not clearly beat it.

Persistence is a hard baseline for monthly field pressure. The Random Forests, which predict the pressure level, lose to it by 9 to 12% because trees cannot extrapolate beyond pressures seen in training; reformulating the target as the monthly change fixes this (Experiment 1). Ridge with drivers then reaches RMSE 82.0 psia against 85.3 for persistence, a 3.9% gain. This gain is not reliable: a paired block bootstrap over the 18 test months gives a 90% interval of -3.4% to +6.4%, MAE is not better (50.3 vs 49.5 psia), and on the 16 months excluding the 2016 shut-in persistence is slightly better (34.9 vs 36.4 psia). A shared five-fold walk-forward CV on the training months, which includes persistence, ranks persistence first (38.2 psia, against 40.6 and 48.4 for the two Ridge variants), mainly because of the first fold, which trains on about 15 months. The honest reading is a tie with persistence.

The LSTM did not change that conclusion. Its seed-to-seed variation is large (for example, RMSE between 69.4 and 86.7 psia across five seeds at LAGS=3), so a single run can beat or lose to persistence by chance. The lowest test RMSE (79.0 psia, LAGS=3, pressure only) was not

selected by training CV, so it is reported as a comparison and not as the final model. The LSTM chosen by CV (LAGS=12, with drivers) scored 90.7 psia, worse than persistence.

Three further experiments produced negative results. Adding production and injection drivers did not help the LSTM on the test months at any lag. Forecasting oil rate instead of pressure was not improved by any model (persistence 0.0506 MMSTB; Ridge 0.0547; LSTMs 0.170 and 0.196). Recursive six-month forecasts tied a flat line (pooled RMSE 127.9 psia for the best model against 132.3 for persistence on the same origins).

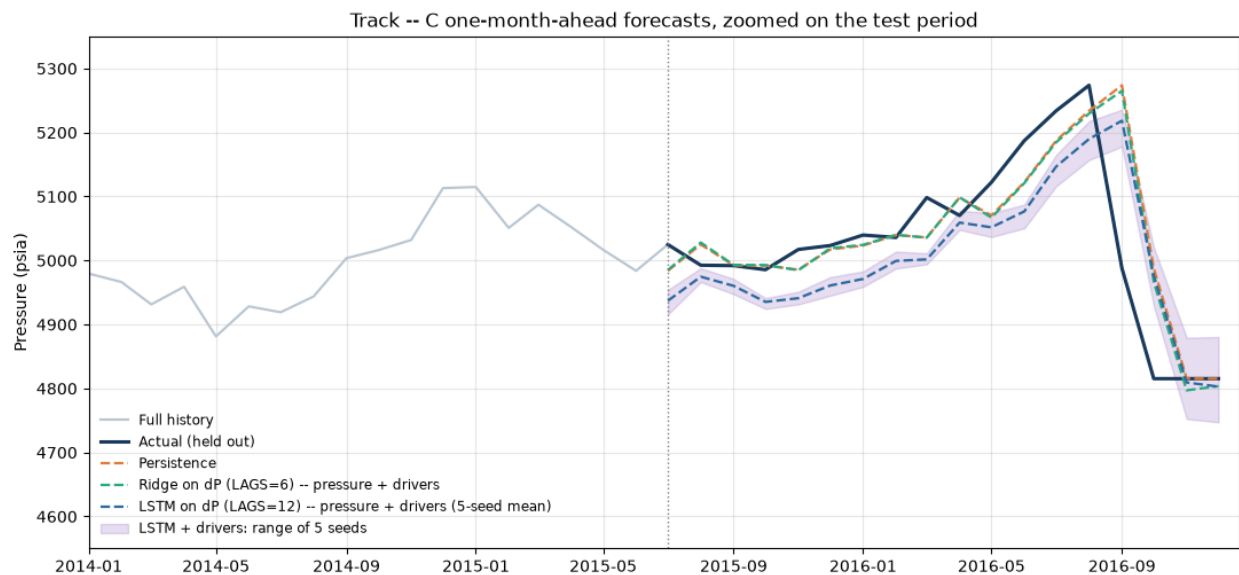


Figure 1 All three models reach the pressure peak one month late and miss the 2016 shut-in.

6. Error Results and Uncertainty

Where the model fails. Errors are concentrated in two months. The Ridge model with drivers misses the 2016-09 and 2016-10 shut-in by +277 and +151 psia, while its next-largest error is 66 psia; these two months carry 82% of its total squared error. Persistence fails on the same months (+286 and +173 psia), so this is a limit of the information in the inputs and not a weakness of one model. The lagged production and injection rates show the slowdown in the months before the shut-in, but the training period contains no shut-in for any model to learn from, so every model can only react one month after the event, as persistence does. Outside the shut-in, Ridge's errors are small (all below about 66 psia) and, on average, slightly negative: it under-forecasts through the 2016 pressure build-up.

Uncertainty. I report an empirical 90% prediction interval. The half-width is the 90th percentile of absolute errors from walk-forward cross-validation on the training months only (70 errors), which gives ± 70 psia. I checked it on the held-out months: it contains 16 of 18 actual pressures (89%), and 100% of the 16 months outside the shut-in. The two months outside the band are exactly the shut-in months, where the assumption that the future resembles the past breaks down. For comparison, persistence has a ± 65 psia interval and covers 15 of 18 months (83%), so the two models have almost the same error band.

The spread across LSTM seeds is a stability check, not an error bar. For the CV-selected LSTM, test RMSE ranges from 88.1 to 99.7 psia across five seeds, yet the seeds agree closely with each other and all miss the shut-in together, so their agreement says nothing about how wrong the forecast is.

On the last 18 months, the Ridge model's one-month-ahead error is about ± 70 psia (90% interval, covering 16 of 18 months); it is reliable for stable operation without sudden rate changes and should not be used to anticipate shut-ins or to forecast more than one month ahead.

7. Limitations and Next Steps

Limitations

- Small sample: 112 monthly points, of which 88 train and 18 test. The result rests on a single 18-month window, and two shut-in months determine most of the test error.
- Field-level data derived from a simulation history match, not raw gauge measurements, so the series is smoother than real pressure data and a model trained on it may not transfer to measured data or to other fields.
- Persistence is not clearly beaten. The best gain (Ridge, 3.9% lower RMSE) is within sampling noise, and a shared training CV ranks persistence first. The LSTM is unstable at this data size, and its results depend on the seed.
- Forecasts are one month ahead. Multi-step forecasts, which hold the drivers constant, did no better than a flat line.
- The prediction interval assumes future months resemble the training months; it failed in the two months that did not.

Next steps

- Add an operational flag for planned shut-ins or rate cuts. The rate decline was visible before the pressure drop, so a known schedule is the most likely way to anticipate the largest errors.
- Evaluate with rolling-origin backtests across the whole history instead of one 18-month window, so that conclusions do not hinge on a single event.
- Test on well-level pressure data (Volve gauge records), and on other fields, to see whether the result holds on measured data.

Appendix: Model Card

Field	Answer
Model name and version	volve-pressure-ridge-dP v1.0: Ridge regression on monthly pressure change, 6-month lags
Intended use	Month-ahead check of field-average pressure for Volve-type reservoirs by reservoir and production engineers, always benchmarked against persistence
Not intended for	Forecasting shut-ins or rate changes; horizons beyond one month; well-level pressure; other fields; replacing a reservoir simulator
Training data	Volve monthly field summary (Equinor open data via yohanesnuwara/pyreservoir); 88 training months, 2008-03 to 2015-06; cumulative volumes differenced to monthly rates (MMSTB)
Inputs and units	Last 5 monthly pressure changes (psia); last 6 months of oil rate and water injection rate (MMSTB/month); 17 features, standardised with training statistics
Output and units	Next-month pressure (psia) = last pressure + predicted change

Evaluation split	Chronological: last 18 months (2015-07 to 2016-12) held out
Headline performance	RMSE 82.0 psia, MAE 50.3 psia vs persistence 85.3 / 49.5. The gain is within noise (90% bootstrap interval -3.4% to $+6.4\%$); persistence ranks first in a shared training CV (38.2 vs 40.6 and 48.4 psia)
Where it fails	2016-09 and 2016-10 shut-in (+277 and +151 psia); excluding those months it is slightly worse than persistence (36.4 vs 34.9 psia)
Uncertainty	Empirical 90% prediction interval ± 70 psia from walk-forward training errors; covers 16/18 test months (89%), 100% excluding the shut-in; persistence ± 65 psia (15/18)
Known limitations and risks	Small sample; simulation-derived field averages; one test window; interval assumes no regime change; LSTM unstable across seeds
How to reproduce	https://github.com/sadaraka-kadi/ml4pe-pressure-forecasting ; Python 3.11.15, numpy 2.4.6, pandas 3.0.3, matplotlib 3.11.0, scikit-learn 1.9.0, torch 2.11.0; run notebooks/capstone.ipynb with Restart & Run All